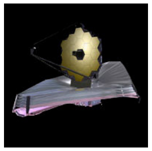

10717 - Testing Fullerene Physics in the HII Regions of C60-rich planetary nebulae

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
SaSt2-3				
	1	MIRI	MIRI Medium Resolution Spectroscopy	(1) PN-SaSt-2-3-MIRI
	2	MIRI Background	MIRI Medium Resolution Spectroscopy	(1) PN-SaSt-2-3-MIRI
	3	NIRSPEC	NIRSpec IFU Spectroscopy	(2) PN-SaSt-2-3-NIRSpec
	4	NIRSPEC Background	NIRSpec IFU Spectroscopy	(3) PN-SaSt-2-3-Background
Lin 49				
	5	MIRI	MIRI Medium Resolution Spectroscopy	(4) LIN-49-MIRI
	6	MIRI Background	MIRI Medium Resolution Spectroscopy	(4) LIN-49-MIRI
	7	NIRSPEC	NIRSpec IFU Spectroscopy	(5) LIN-49-NIRSpec
	8	NIRSPEC Background	NIRSpec IFU Spectroscopy	(6) LIN-49-Background

ABSTRACT

Recent JWST/MIRI observations of the planetary nebula Tc 1 have revealed that C60 exists as gas-phase molecules residing within the HII region and not in a photo-dissociation region as generally assumed. Photodetachment of C60- provides a shielding mechanism that allows survival in such harsh UV radiation fields. These findings challenge fundamental assumptions about the molecular physics and physical state of fullerenes (and other large aromatic molecules) in space.

We propose JWST/MIRI-MRS and NIRSpec-IFU observations of SaSt2-3 and Lin 49, the only two remaining accessible "pure" fullerene planetary nebulae. These targets have similar stellar temperatures to Tc 1 but substantially different radiation environments, as indicated by their [Ar II]/[Ar III] line ratios. This systematic comparison will determine whether gas-phase C60 in HII regions represents general fullerene physics or requires specific G0/n_e conditions. Our spatially resolved spectroscopy will: (1) test for gas-phase C60 via ro-vibrational structure; (2) determine where C60 resides relative to the ionization structure; (3) characterize how G0/n_e affects C60 excitation, charge state, and spatial distribution; (4) determine whether the photodetachment mechanism operates broadly in HII regions.

The results of this program will establish whether astrophysical models incorporating aromatic molecule physics require fundamental revision, with implications extending from molecular survival and excitation mechanisms to gas heating processes throughout star-forming environments.

OBSERVING DESCRIPTION

We will observe the C60-rich planetary nebulae SaSt2-3 and Lin 49 to study how C60 and other fullerenes respond to changes in their physical environment and to determine whether the recent JWST findings about C60 in Tc 1 are universal. To spatially and spectroscopically resolve the fullerene features in these different environments, we request 2.87-5.27 micron NIRSpec IFU at high spectral resolution and 5-28 micron MIRI/MRS observations of both targets.

For NIRSpec, we use the G395H/F290LP disperser-filter combination, the NRSIRS2RAPID readout pattern for Lin 49 and the NRSIRS2 readout pattern for SaSt2-3, and a 4-point dither pattern. We also obtain Leakcal observations and dedicated background observations.

For MIRI/MRS, we use the FASTR1 readout pattern and the “Extended Source, 4-Point ALL" dither pattern. Following JWST documentation recommendations, we perform MIRI MRS simultaneous imaging to improve the astrometric solution of individual MRS exposures. Dedicated background observations are also obtained with MIRI MRS simultaneous imaging of the target.

As our targets are extended, we do not need target acquisition.

Since the background signal varies with time for a given position, we request for non-interruptible observations of the science pointings and their corresponding dedicated background pointings. We set a PA offset requirement between the NIRSpec and MIRI observation of each given target to maximize the overlap between the NIRSpec and MIRI FOV. We add a PA requirement to the MIRI observations so that the NIRSpec and MIRI observations (tied through the PA offset requirement) both occur in the meteoroid safe visibility window. The PA requirement for Lin 49 is then further constrained such that the background observations do not contain any stars in the FOV.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	PN-SaSt-2-3-MIRI	RA: 07 48 3.6733 (117.0153054d) Dec: -14 07 40.43 (-14.12790d) Equinox: J2000	Proper Motion RA: -0.597 mas/yr Proper Motion Dec: 0.745 mas/yr Parallax: 8.649999999999999E-5" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Planetary nebulae] Extended=YES				
(2)	PN-SaSt-2-3-NIRSpec	RA: 07 48 3.6733 (117.0153054d) Dec: -14 07 40.43 (-14.12790d) Equinox: J2000	Proper Motion RA: -0.597 mas/yr Proper Motion Dec: 0.745 mas/yr Parallax: 8.649999999999999E-5" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Planetary nebulae] Extended=YES				
(3)	PN-SaSt-2-3-Background	RA: 07 48 3.4596 (117.0144150d) Dec: -14 08 26.13 (-14.14059d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=ISM Description=[Planetary nebulae] Extended=YES				
(4)	LIN-49-MIRI	RA: 00 43 53.8870 (10.9745292d) Dec: -72 55 14.79 (-72.92077d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Planetary nebulae] Extended=YES				
(5)	LIN-49-NIRSpec	RA: 00 43 53.8870 (10.9745292d) Dec: -72 55 14.79 (-72.92077d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Planetary nebulae] Extended=YES				
(6)	LIN-49-Background	RA: 00 43 55.4500 (10.9810417d) Dec: -72 54 50.71 (-72.91409d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Planetary nebulae] Extended=YES				

Observation

Proposal 10717, Observation 1: MIRI

Fri Mar 13 20:00:30 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Data Excess over lower threshold

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	PN-SaSt-2-3-MIRI	RA: 07 48 3.6733 (117.0153054d) Dec: -14 07 40.43 (-14.12790d) Equinox: J2000	Proper Motion RA: -0.597 mas/yr Proper Motion Dec: 0.745 mas/yr Parallax: 8.649999999999999E-5" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	5	1	1	Dither 1	4	4	55.501	
1	SHORT(A)	MRSLONG		FASTR1	22	14	1	Dither 1	4	56	3563.151	
1	SHORT(A)	MRSSHORT		FASTR1	175	2	1	Dither 1	4	8	3896.156	
2		IMAGER	F1000W	FASTR1	5	1	1	Dither 1	4	4	55.501	
2	MEDIUM(B)	MRSLONG		FASTR1	165	2	1	Dither 1	4	8	3674.153	
2	MEDIUM(B)	MRSSHORT		FASTR1	180	2	1	Dither 1	4	8	4007.158	
3		IMAGER	F1130W	FASTR1	5	1	1	Dither 1	4	4	55.501	
3	LONG(C)	MRSLONG		FASTR1	150	2	1	Dither 1	4	8	3341.148	
3	LONG(C)	MRSSHORT		FASTR1	150	2	1	Dither 1	4	8	3341.148	

Special Requirements	
	Aperture PA Range 100.0 to 102.0 Degrees (V3 100.0 to 102.0) Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 10717, Observation 2: MIRI Background

Fri Mar 13 20:00:30 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Data Excess over lower threshold

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	PN-SaSt-2-3-MIRI	RA: 07 48 3.6733 (117.0153054d) Dec: -14 07 40.43 (-14.12790d) Equinox: J2000	Proper Motion RA: -0.597 mas/yr Proper Motion Dec: 0.745 mas/yr Parallax: 8.649999999999999E-5" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	Imager	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	5	30	1	Dither 1	4	120	1986.929	
1	SHORT(A)	MRSLONG		FASTR1	22	14	1	Dither 1	4	56	3563.151	
1	SHORT(A)	MRSSHORT		FASTR1	175	2	1	Dither 1	4	8	3896.156	
2		IMAGER	F1000W	FASTR1	5	30	1	Dither 1	4	120	1986.929	
2	MEDIUM(B)	MRSLONG		FASTR1	165	2	1	Dither 1	4	8	3674.153	
2	MEDIUM(B)	MRSSHORT		FASTR1	180	2	1	Dither 1	4	8	4007.158	
3		IMAGER	F1130W	FASTR1	5	30	1	Dither 1	4	120	1986.929	
3	LONG(C)	MRSLONG		FASTR1	150	2	1	Dither 1	4	8	3341.148	
3	LONG(C)	MRSSHORT		FASTR1	150	2	1	Dither 1	4	8	3341.148	

Special Requirements	Sequence Observations 1, 2, Non-interruptible

Observation

Proposal 10717, Observation 3: NIRSPEC

Fri Mar 13 20:00:30 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observations:[NIRSPEC Background (Obs 4)]

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	PN-SaSt-2-3-NIRSpec	RA: 07 48 3.6733 (117.0153054d) Dec: -14 07 40.43 (-14.12790d) Equinox: J2000	Proper Motion RA: -0.597 mas/yr Proper Motion Dec: 0.745 mas/yr Parallax: 8.649999999999999E-5" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSIRS2	10	5	false	true	NONE	4	20	14880.668	
2	G395H/F290LP	NRSIRS2	10	5	true	true	NONE	4	20	14880.668	

Special Requirements	
	Aperture PA Range 197.97164917 to 199.97164917 Degrees (V3 59.0 to 61.0) Sequence Observations 3, 4, Non-interruptible

Observation

Proposal 10717, Observation 4: NIRSPEC Background

Fri Mar 13 20:00:30 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observation For: [NIRSPEC (Obs 3)]

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	PN-SaSt-2-3-Background	RA: 07 48 3.4596 (117.0144150d) Dec: -14 08 26.13 (-14.14059d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSIRS2	10	5	false	true	NONE	4	20	14880.668	

Special Requirements

Sequence Observations 3, 4, Non-interruptible

Observation

Proposal 10717, Observation 5: MIRI

Fri Mar 13 20:00:30 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(MIRI (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	LIN-49-MIRI	RA: 00 43 53.8870 (10.9745292d) Dec: -72 55 14.79 (-72.92077d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	5	1	1	Dither 1	4	4	55.501	
1	LONG(C)	MRSLONG		FASTR1	25	2	1	Dither 1	4	8	566.108	
1	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
2		IMAGER	F770W	FASTR1	5	1	1	Dither 1	4	4	55.501	
2	MEDIUM(B)	MRSLONG		FASTR1	28	4	1	Dither 1	4	16	1276.518	
2	MEDIUM(B)	MRSSHORT		FASTR1	15	8	1	Dither 1	4	32	1409.72	
3		IMAGER	F1000W	FASTR1	5	1	1	Dither 1	4	4	55.501	
3	SHORT(A)	MRSLONG		FASTR1	10	5	1	Dither 1	4	20	599.409	
3	SHORT(A)	MRSSHORT		FASTR1	25	2	1	Dither 1	4	8	566.108	

Special Requirements	
	Aperture PA Range 334.3 to 339.4 Degrees (V3 334.3 to 339.4) Sequence Observations 5, 6, Non-interruptible V3 PA Offset 5 from 7 by 48 to 50 Degrees (Aperture -90.97164917 to -88.97164917)

Observation

Proposal 10717, Observation 6: MIRI Background

Fri Mar 13 20:00:30 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	LIN-49-MIRI	RA: 00 43 53.8870 (10.9745292d) Dec: -72 55 14.79 (-72.92077d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	Imager	YES	BRIGHTSKY	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	5	20	1	Dither 1	4	80	411.873	
1	LONG(C)	MRSLONG		FASTR1	25	2	1	Dither 1	4	8	566.108	
1	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
2		IMAGER	F770W	FASTR1	5	20	1	Dither 1	4	80	411.873	
2	MEDIUM(B)	MRSLONG		FASTR1	28	4	1	Dither 1	4	16	1276.518	
2	MEDIUM(B)	MRSSHORT		FASTR1	15	8	1	Dither 1	4	32	1409.72	
3		IMAGER	F1000W	FASTR1	5	20	1	Dither 1	4	80	411.873	
3	SHORT(A)	MRSLONG		FASTR1	10	5	1	Dither 1	4	20	599.409	
3	SHORT(A)	MRSSHORT		FASTR1	25	2	1	Dither 1	4	8	566.108	

Special Requirements	
	Offset -5.0 arcsec, -30.0 arcsec Sequence Observations 5, 6, Non-interruptible

Observation

Proposal 10717, Observation 7: NIRSPEC

Fri Mar 13 20:00:30 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observations:[NIRSPEC Background (Obs 8)]

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(NIRSPEC (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	LIN-49-NIRSpec	RA: 00 43 53.8870 (10.9745292d) Dec: -72 55 14.79 (-72.92077d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSIRS2RAPID	5	6	false	true	NONE	4	24	2100.8	
2	G395H/F290LP	NRSIRS2RAPID	5	6	true	true	NONE	4	24	2100.8	

Special Requirements

Sequence Observations 7, 8, Non-interruptible
V3 PA Offset 5 from 7 by 48 to 50 Degrees (Aperture -90.97164917 to -88.97164917)

Observation

Proposal 10717, Observation 8: NIRSPEC Background

Fri Mar 13 20:00:30 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observation For: [NIRSPEC (Obs 7)]

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	LIN-49-Background	RA: 00 43 55.4500 (10.9810417d) Dec: -72 54 50.71 (-72.91409d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=ISM
Description=[Planetary nebulae]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	NRSIRS2RAPID	5	6	false	true	NONE	4	24	2100.8	

Special Requirements

Sequence Observations 7, 8, Non-interruptible